CHAPTER 4

# Random Numbers: Fake Dice, Real Power



System Simulation & Modeling · Master 1 RSD

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# Warm-up: can a computer be random?

Yes, always

No, never

It fakes it so well nobody notices



**Vote with your hand!** 🙋

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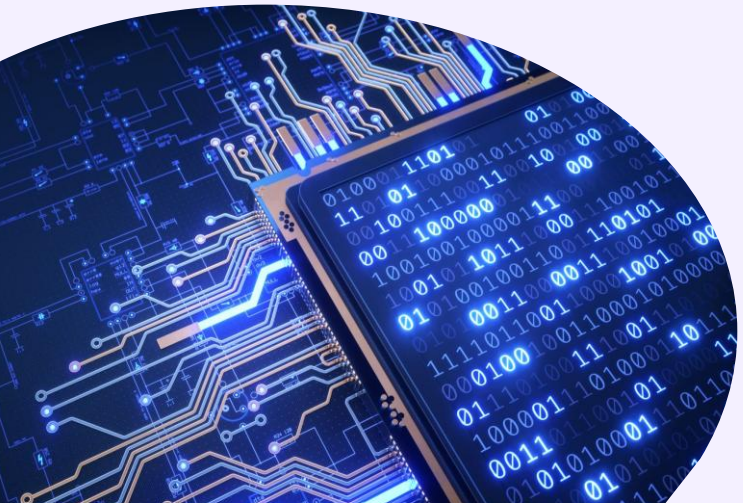
# Why simulations need dice

## IT
Packets reach a router at random moments. Is the buffer big enough?

## Healthcare
Patients walk into the ER whenever they want. How many nurses do we need?

## Manufacturing
A machine breaks down… nobody knows when. How many spare parts?

*No randomness = a boring simulation that never surprises us.*







# Ask a computer for a random number…

Hey computer, give me a random number!



Sure! 4. Again? 4… Wait, same start = same numbers. Oops.



# True random vs pseudo-random

## True random 🎲

Comes from physical noise: dice, radioactive decay, sensors.

✓ Unpredictable
✗ Slow
✗ Can’t replay it

## Pseudo-random 🤖

A formula + a starting point (the seed).

✓ Fast
✓ Replayable
✓ Looks random enough

# What makes a generator good?

1

## Long period

A playlist that doesn’t repeat for years

2

## Uniform

A fair dice: every value equally likely

3

## Independent

Last number tells you nothing about the next

# The seed: your replay button

## Same seed → same numbers

Why it's great:
• Debugging: reproduce the exact bug
• Fair comparison: test 2 hospital staffing plans on the same "day"
• Science: others can repeat your results

**Bug in run #37!** 😱

**Use seed 12345 and we see it again.**

# Meet the LCG: one tiny formula

$$X(next) = (a \times X + c) \bmod m \qquad U = X / m$$

**$X_0$**
seed: where we start

**a**
multiplier

**c**
increment

**m**
modulo: the clock size

*“mod” = go around the clock and come back to 0.*

# By hand: seed 7, a=5, c=3, m=16

**X values**

| | | | | | |
|---|---|---|---|---|---|
| 7 | 6 | 1 | 8 | 11 | 10 |

**U = X / 16**

| | | | | | |
|---|---|---|---|---|---|
| 0.44 | 0.38 | 0.06 | 0.50 | 0.69 | 0.63 |

First step: 5 × 7 + 3 = 38, and 38 mod 16 = 6. After 16 numbers we are back to 7: period = 16 (the maximum!)



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# Your turn! ⏱ 5 minutes

**Seed = 10, a = 5, c = 3, m = 16**

1. Compute $X_1$ and $X_2$
2. Convert them to U values
3. Bonus: find $X_3$

**Calculators allowed. Brains encouraged.**



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# Oops: a bad choice of parameters

**m = 10, a = 3, c = 1, seed = 7**

7 2 7 2 7 2

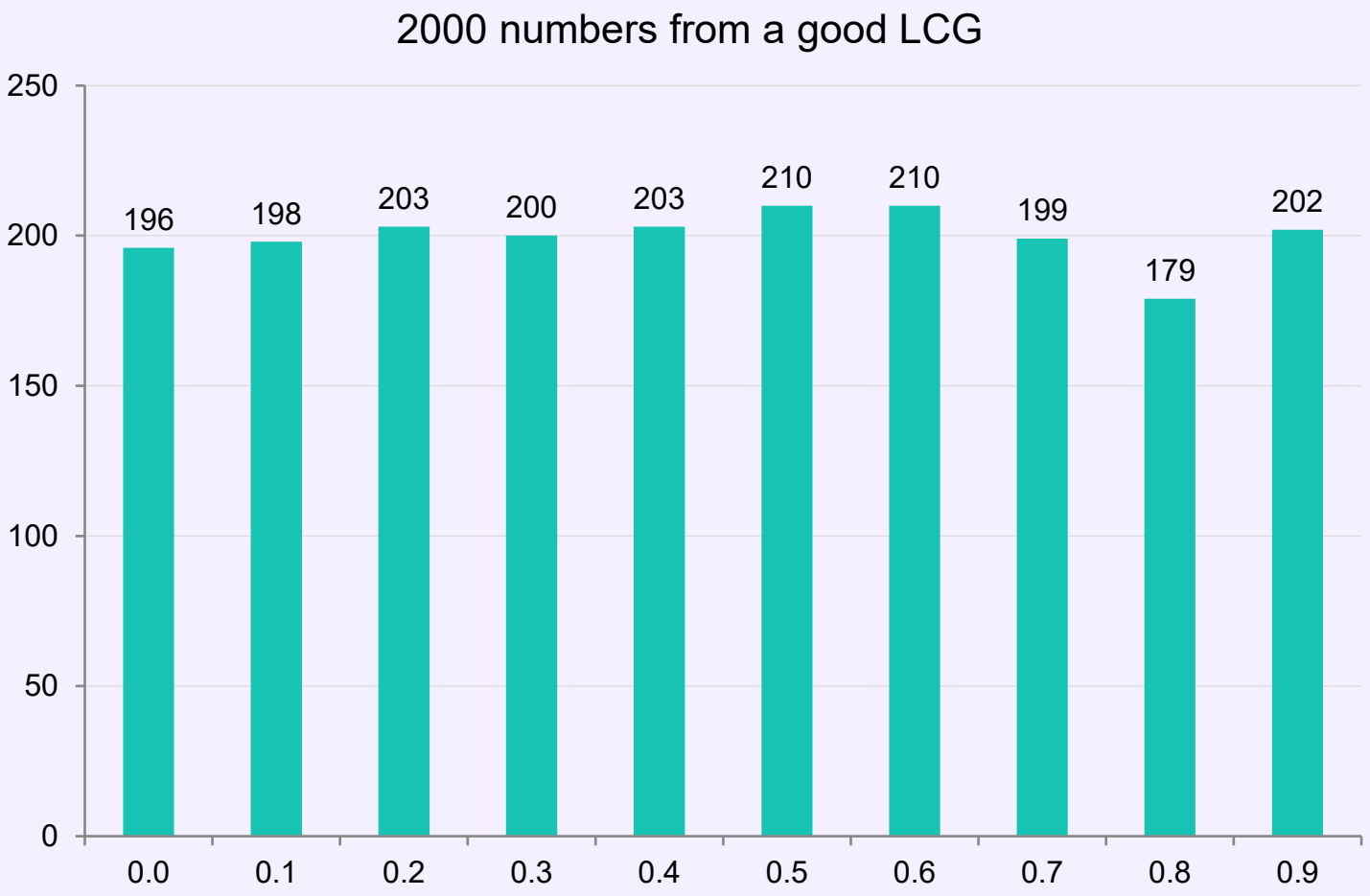
## Period = 2 🔁

Our “random” machine breakdown happens at the same two moments forever.

## Rule of thumb

Use a big m. Example: m = $2^{32}$ gives about 4 billion numbers before repeating.

# Test 1: is it fair? (histogram)

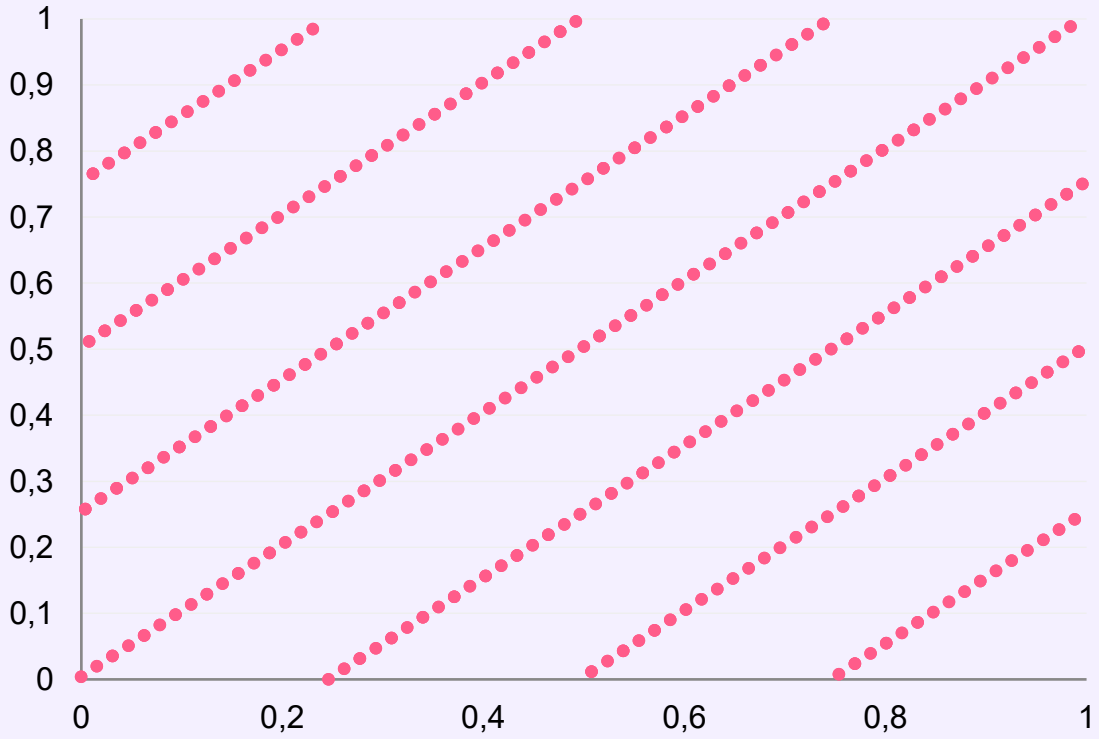



**Flat bars ≈ fair dice**
About 200 per bar expected.

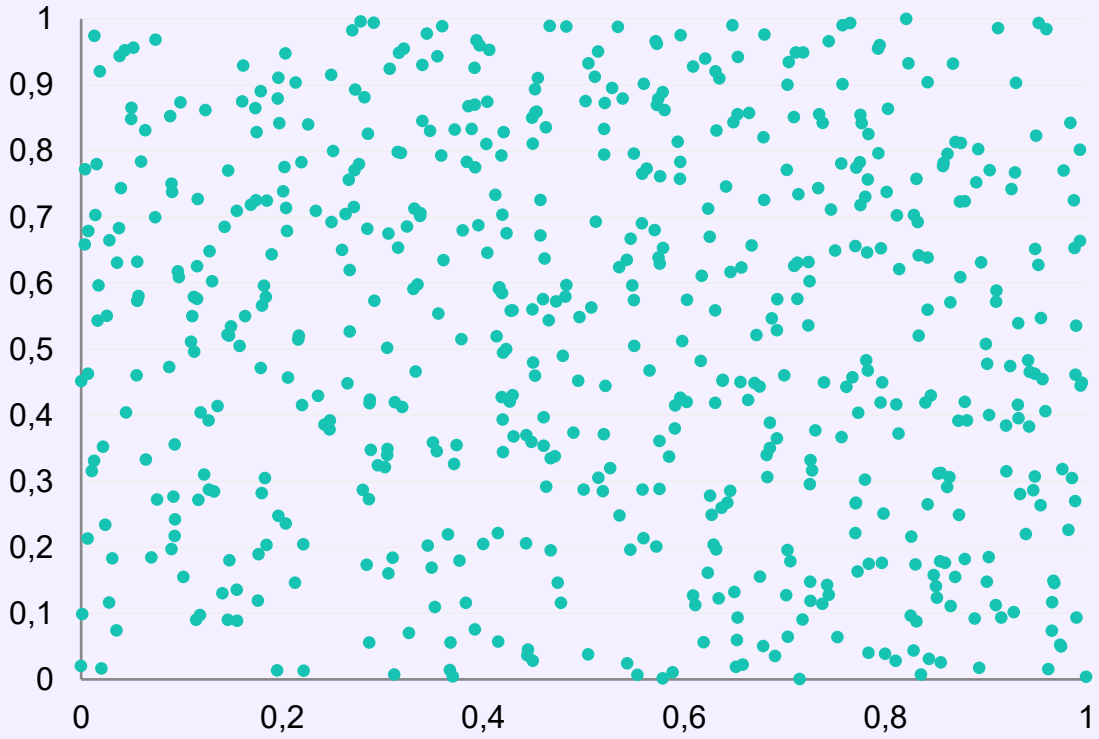
Is flat enough to say “good”? 😟
Next slide…

# Test 2: independent? Plot ($U_n$, $U_{n+1}$)

Small LCG (m = 256): lines ✗




Big LCG (m = $2^{32}$): cloud ✓




**Same uniform histogram, very different quality!**

# Bad randomness = bad network

Collision! I'll wait 3 slots.

Me too, 3 slots!

Collision again… and again…

# The birthday surprise



**~23**

people in a room: 50% chance two share a birthday

**~300**

devices picking random IPs among 65,536: 50% chance of a duplicate

**~120**

random 4-digit patient IDs (10,000 possible): 50% chance of a duplicate

*Randomness is not the same as “never repeats”.*

# When guessable = hackable

## DNS with a 16-bit ID

An attacker sprays fake answers hoping to guess the ID.

A few hundred packets can be enough, in under a second.

## Add a random port

Now the guess space is about 65,000 times bigger.

Same idea, much harder attack. More random bits = more safety.

# From U to real life

**Waiting time = − (average gap) × ln(U)**

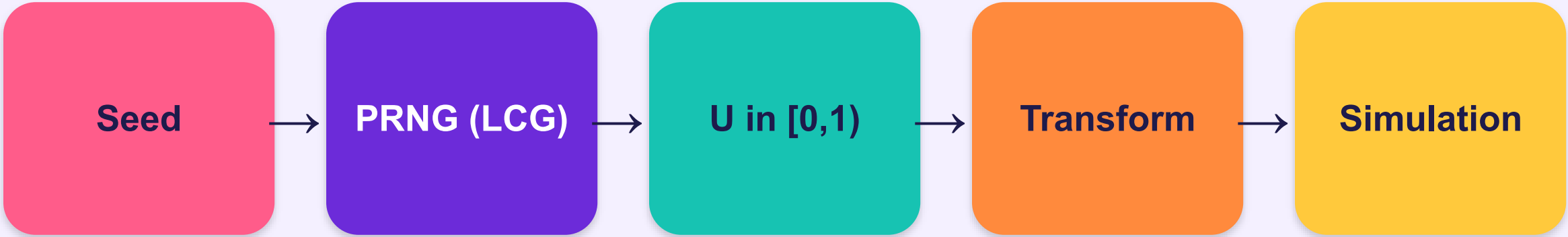
**ER example: 6 patients per hour → average gap = 10 min**

**U = 0.90**
**≈ 1 min**
Rush hour!

**U = 0.50**
**≈ 7 min**
Typical

**U = 0.05**
**≈ 30 min**
Quiet night

# The whole story in one line




*Garbage numbers in → garbage conclusions out.*

# Quiz time! 🧠 (10 min)

1. Run the same seed twice: same numbers or different?

2. LCG with m = 16 or m = $2^{32}$: which one for a real simulation?

3. Why fix the seed when debugging a hospital simulation?

4. 300 random IPs among 65,536: why do collisions appear so early?

# Next: TP1, your turn to build it

**1**
Write lcg(seed, n, m, a, c)

**2**
Compare 3 LCGs with plots

**3**
Random IPs: how many collisions?

**4**
Attack a DNS ID with Kaminsky

**Rule: no ready-made random generators from libraries!**

# Take-home

**Computers fake randomness with a formula and a seed**

**Good = long period, uniform, independent**

**Same seed = replayable simulations**

**Always check with plots, not just histograms**

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